

Solution Conformation of Proteinase Inhibitor IIA from Bull Seminal Plasma by ^1H Nuclear Magnetic Resonance and Distance Geometry

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A determination of the solution conformation of the proteinase inhibitor IIA from bull seminal plasma (BUSI IIA) is described. Two-dimensional nuclear Overhauser enhancement spectroscopy (NOESY) was used to obtain a list of 202 distance constraints between individually assigned hydrogen atoms of the polypeptide chain, to identify the positions of the three disulfide bridges, and to locate the single *cis* peptide bond. Supplementary geometric constraints were derived from the vicinal spin–spin couplings and the locations of certain hydrogen bonds, as determined by nuclear magnetic resonance (n.m.r.). Using a new distance geometry program (DISGEO) which is capable of computing all-atom structures for proteins the size of BUSI IIA, five conformers were computed from the NOE distance constraints alone, and another five were computed with the supplementary constraints included. Comparison of the different structures computed from the n.m.r. data among themselves and with the crystal structures of two homologous proteins shows that the global features of the conformation of BUSI IIA (i.e. the overall dimensions of the molecule and the threading of the polypeptide chain) were well-defined by the available n.m.r. data. In the Appendix, we describe a preliminary energy refinement of the structure, which showed that the constraints derived from the n.m.r. data are compatible with a low energy spatial structure.

1. Introduction

This paper describes the determination of the solution conformation of proteinase inhibitor IIA from bull seminal plasma in aqueous solution. BUSI IIA† is a protein of 57 amino acid residues and with a molecular weight of 6000, and was first described by Čechová *et al.* (1979). It inhibits proteinases such as trypsin, and may play a part in the protection of tissues against inflammatory processes (Fritz *et al.*, 1978), and possibly also in the control of fertilization (Schiessler *et al.*, 1976).

The structure determination described here relies on the use of nuclear magnetic resonance to derive a list of proton–proton distance constraints, followed by application of a distance geometry program, DISGEO (Havel & Wüthrich, 1984), to compute structures consistent with those constraints. In the preceding paper (Havel & Wüthrich, 1985), the

DISGEO program was evaluated using n.m.r. data which had been simulated from the known crystal structure of BPTI. Comparison of the results of

† Abbreviations used: BUSI IIA, proteinase inhibitor IIA from bull seminal plasma; BPTI, basic pancreatic trypsin inhibitor; PSTI, porcine secretory trypsin inhibitor; OMJPQ3, third domain of the Japanese quail ovomucoid; n.m.r., nuclear magnetic resonance; NOE, nuclear Overhauser enhancement; NOESY, 2-dimensional nuclear Overhauser enhancement spectroscopy; COSY, 2-dimensional correlated spectroscopy; $^3J_{\text{HNH}}$, vicinal spin–spin coupling constant between the amide proton and the C $^{\alpha}$ proton of the same amino acid residue, in units of Hertz (Hz); p.p.m., parts per million; DISGEO, name of distance geometry program; RMS, root-mean-square; RMSD, root-mean-square co-ordinate difference; DHAD, dihedral angle difference.

these computations with the crystal structure provided an estimate of how well a protein spatial structure can be determined from the geometric constraints which are available from n.m.r. These results are used here to assess the quality of the solution conformations which we have computed for BUSI IIA, for which no crystal structure is available. Further evidence for the correctness of the structure determined by n.m.r. is obtained by comparison with the crystal structures of two homologous proteins, porcine secretory inhibitor (Bolognesi *et al.*, 1982), and the third domain of the Japanese quail ovomucoid (Papamokos *et al.*, 1982).

The structural features of BUSI IIA elucidated by this study are the direct result of distance geometry calculations using geometric constraints derived exclusively from its primary structure and from the n.m.r. experiments. In the Appendix, we describe preliminary results from an energy minimization by which the side-chain packing and local structure were further improved, and which demonstrates the existence of low energy conformations that are consistent with the experimental data.

2. Materials and Methods

(a) n.m.r. measurements

BUSI IIA, purified by the method of Čechová *et al.* (1979), was obtained as a gift from Drs D. Čechová and P. Štřop. For the n.m.r. experiments, approx. 0.016 M solutions of BUSI IIA were prepared, both in $^2\text{H}_2\text{O}$ and in a mixture of 90% H_2O and 10% $^2\text{H}_2\text{O}$, at pH 5.3. No buffer was used, and the spectra were run at 45°C. For the $^2\text{H}_2\text{O}$ spectra, the labile protons of the protein were replaced by ^2H by heating the solution to 40°C for 20 min at neutral p ^2H , and the residual water protons were removed by lyophilization from $^2\text{H}_2\text{O}$.

NOESY spectra (Anil Kumar *et al.*, 1980) were recorded at 500 MHz on a Bruker WM 500 spectrometer equipped with an Aspect 2000 computer and a Control Data Corp. disk drive. To reduce contributions from coherent magnetization transfer to the cross peak intensities, mixing times shorter than 200 ms were randomly modulated with a modulation amplitude of ± 5 to 10 ms, depending on the mixing time (Macura *et al.*, 1981). For measurements in H_2O the solvent resonance was suppressed by selective, continuous irradiation at all times except during data acquisition (Wider *et al.*, 1983). Absorption mode spectra were obtained as described (Williamson *et al.*, 1984), using the time proportional phase incrementation scheme (Redfield & Kunz, 1975; Bodenhausen *et al.*, 1980; Marion & Wüthrich, 1983).

A high resolution NOESY spectrum in H_2O was acquired with 4096 points in t_2 and 800 points in t_1 , with a mixing time τ_m of 200 ms. It was processed as described by Williamson *et al.* (1984), to give a digital resolution of 2.4 Hz in ω_1 and 0.6 Hz in ω_2 . Lower resolution NOESY spectra with τ_m values of 50 ms, 80 ms, 100 ms and 120 ms had 1024 points in t_2 and 512 points in t_1 . The time domain data matrix was expanded to 1024 points in t_1 by zero filling, and the window function used in both directions was a Lorentz-to-Gaussian transformation with a line-broadening factor of -1 and a maximum at 0.3 t_2 . This window function was used as it was felt to be the least discriminatory for different linewidths, consistent

with eliminating most of the t_1 noise and the dispersion parts of slightly incorrectly phased peaks (Ferrige & Lindon, 1978). Similarly, a high resolution NOESY spectrum with $\tau_m = 150$ ms and several lower resolution spectra with shorter mixing times were acquired in $^2\text{H}_2\text{O}$.

(b) ^1H - ^1H distance constraints derived from NOESY spectra

In the high resolution NOESY spectra of BUSI IIA recorded in H_2O and in $^2\text{H}_2\text{O}$, 556 cross peaks were assigned with the help of the previously reported chemical shifts (Štřop *et al.*, 1983). Distance constraints were then derived from 202 of these cross peaks, using NOESY spectra recorded with lower digital resolution and a shorter mixing time (τ_m) in order to avoid spin diffusion (Kalk & Berendsen, 1976; Gordon & Wüthrich, 1978; Anil Kumar *et al.*, 1981). Among the assigned cross peaks that were not used for the structure determination are the majority of the intrasidue NOEs. Many of these manifest ^1H - ^1H distances which are fixed by the covalent structure, and others are not informative at the present level of quantification. Most of the remaining unused NOEs are sequential NOEs between C^βH_i and NH_{i+1} . They were not used because the absence of stereospecific assignments (Wüthrich *et al.*, 1983) made them uninformative. Roughly 20 additional NOEs could be resolved but not assigned; by comparing these to expected close proton-proton contacts in the computed structures, we conclude that most of them probably arise from unassigned C^β and C^γ protons.

For the calibration of correlations between NOESY cross peak heights and ^1H - ^1H distance constraints we proceeded in a manner similar to that used by Braun *et al.* (1983). Provided that a specified cross peak could be seen at high digital resolution to be sufficiently well-resolved from all other peaks, the height at the centre of this peak was measured in the spectra recorded with $\tau_m = 50$ ms, 80 ms, 100 ms and 120 ms, and the peak heights plotted *versus* τ_m . To check that the observed peak intensities were due to direct NOEs (Anil Kumar *et al.*, 1981), a smooth curve was drawn through these points. Subsequently, the NOE build-up was assumed to be linear up to 80 ms, and the peak heights at this value of τ_m were used to determine relative initial NOE build-up rates for the individual pairs of dipolar-coupled protons.

For connectivities between protons separated by more than 3 single bonds in the covalent structure, differences between the apparent build-up rates for different pairs of protons were not further interpreted. All such 'long-range' connectivities between backbone amide and/or C^α protons which could be shown to arise from direct NOEs and which had a sufficiently high intensity to be unambiguously identified in the spectrum with $\tau_m = 80$ ms were attributed a distance range from the van der Waals' contact of 2.0 Å to an upper bound of 4.0 Å. Similarly, long-range connectivities which also involved side-chain protons were attributed a distance range from 2.0 to 5.0 Å. The different upper bounds for these 2 classes of constraints were chosen on the basis of the implications from a uniform averaging model (Braun *et al.*, 1981) for the influence of internal mobility on the observed NOESY cross peaks, assuming a somewhat higher overall flexibility for the amino acid side-chains than for the backbone segments.

For the intrasidue and sequential connectivities between amide, C^α and C^β protons with at most 3 intervening torsion angles about C-C or C-N single

bonds, a semiquantitative relation between NOESY cross peak intensities and upper bounds on the corresponding distances was established on the basis of empirical calibrations using the following known sequential distances (Billeter *et al.*, 1982): $d_{\alpha\text{N}} = 2.3$ Å in β -sheets; $d_{\text{NN}} = 2.8$ Å in α or 3_{10} helices; $(d_{\alpha\text{N}})_{\text{max}} = 3.6$ Å; $d_{\text{NN}} = 4.3$ Å in β -sheets. For example, all cross peaks corresponding to $d_{\alpha\text{N}}$ distances in β -sheets (2.3 Å) had a relative intensity of greater than 150. An upper bound of 2.5 Å was therefore set for all cross peaks of intensity greater than 150. The resulting calibration of the relative cross peak intensities *versus* upper bound constraints is given in Table 1.

NOEs to hydrogen atoms which could not be stereospecifically assigned were referred to pseudo atom positions, using the previously reported corrections for the experimental NOE constraints (Wüthrich *et al.*, 1983).

(c) *Additional geometric constraints derived from NOESY and other n.m.r. experiments*

As described in Results, the location of the 3 disulfide bonds in BUSI IIA could be unambiguously identified from the observed NOEs. The presence of these disulfide bonds was then fixed directly by imposing a range of 2.0 to 2.1 Å on the S-S distance, and of 3.0 to 3.1 Å on the S-C $^{\beta}$ distances across each bridge.

The locations of backbone amide proton to carbonyl oxygen hydrogen bonds in certain positions of the regular secondary structures could be inferred from the observed NOEs and the slow amide proton exchange rates (Wagner *et al.*, 1981; Williamson *et al.*, 1984). These hydrogen bonds were defined by specifying a range of 1.8 to 2.0 Å for the H-O distance, and 2.7 to 3.0 Å for the N-O distance, which forces the hydrogen bonds to be approximately linear. Other hydrogen bonds between the backbone amide proton and the γ -carbonyl oxygen of certain glutamic acid residues were identified from pH titrations of the amide proton chemical shifts (Bundi & Wüthrich, 1979; Ebina & Wüthrich, 1984). These were fixed by imposing the same constraints as used for the backbone hydrogen bonds on the distances between the amide proton, and the nitrogen and carbonyl oxygen atoms.

Table 1

Calculated correlations between intensity of the NOESY cross peaks manifesting sequential or intra-residue connectivities between amide, C $^{\alpha}$ and C $^{\beta}$ protons in BUSI IIA and upper bounds on the ^1H - ^1H distance constraints

Protons connected†	$I_{\text{rel}}^{\ddagger}$	Upper distance bound§ (Å)
NH, C $^{\alpha}$ H	>150	2.5
NH, C $^{\beta}$ H	60-150	3.0
C $^{\alpha}$ H, C $^{\beta}$ H	20-60	4.0
NH, NH	>100	3.0
	20-100	4.0

† Different calibrations were used for cross peaks involving only amide protons and those involving C $^{\alpha}$ H or C $^{\beta}$ H, since these have characteristically different fine structures.

‡ Relative cross peak heights in the NOESY spectra recorded with a mixing time of 80 ms, in arbitrary units.

§ The lower bound on the distance range is in all cases the van der Waals' distance of 2.0 Å.

Spin-spin coupling constants were measured in phase-sensitive, high resolution COSY spectra (Marion & Wüthrich, 1983; Williamson *et al.*, 1984). An unambiguous interpretation in terms of the dihedral angle can be made only for extremely small or extremely large couplings (Pardi *et al.*, 1985). Accordingly, the ϕ angles corresponding to $^3J_{\text{HN}\alpha} \geq 8.0$ Hz were constrained in the range of -160° to -80° , and the ϕ angles corresponding to $^3J_{\text{HN}\alpha} \leq 5.5$ Hz were constrained in the range of -90° to -40° . In addition, the χ^1 angles of Val, Ile and Thr corresponding to $^3J_{\alpha\beta} \geq 8.0$ Hz were confined to a range of $\pm 30^\circ$ about the *trans* position of C $^{\alpha}$ H and C $^{\beta}$ H. These dihedral angle ranges were imposed using chirality constraints, as described by Havel & Wüthrich (1985).

(d) *Geometric constraints derived from the primary structure*

Besides the n.m.r. constraints described above, the constraints imposed by the primary structure of the polypeptide play an important role in the distance geometry calculations. In the present work, the ECEPP amino acid geometries (Némethy *et al.*, 1983) were used to obtain the bond lengths and geminal distances which determine bond angles. Chirality constraints (Havel *et al.*, 1983; Havel & Wüthrich, 1985) were used to fix the chiralities of the amino acids and the planarity of all sp^2 centers. The atomic hard sphere radii described by Wüthrich *et al.* (1983) were used for all atoms, except between pairs of vicinal atoms, where the van der Waals' repulsion was ignored to facilitate the convergence of the distance geometry program (Havel *et al.*, 1983; Havel & Wüthrich, 1985). The total number of constraints thus obtained exceeds by far the minimum number of $3N-6$ constraints needed to define a structure. However, as discussed by Havel & Wüthrich (1985), the exact number of constraints available cannot be related in any simple way to the precision with which the structure is determined.

(e) *Analysis of the computed structures*

For the calculation of the spatial molecular structure, the DISGEO distance geometry program (Havel & Wüthrich, 1984) was used with 2 different sets of experimental input data. The calculations were repeated 5 times for each of these data sets, using different initial values for the random number seed, to obtain a total of 10 different conformers each consistent with its respective input data. These conformers were then compared with one another in a variety of ways in order to decide to what extent specific structural features were uniquely determined by the experimental data (Havel *et al.*, 1979; Braun *et al.*, 1983; Havel & Wüthrich, 1985).

First, the global arrangements of the polypeptide chain in the 5 conformers were compared using the root-mean-square co-ordinate deviation (RMSD) between all pairs of spatial structures. The RMSD is given by the minimum over all rotations about the superimposed centroids, $\mathbb{R}$, of the RMS difference in the n atomic co-ordinates between the 2 conformers (Nyburg, 1974):

$$\text{RMSD} = \left[\frac{1}{n} \min_{\mathbb{R}} \sum_{i=1}^n \|\vec{r}_i - \mathbb{R} \vec{r}'_i\|^2 \right]^{1/2}, \quad (1)$$

where $\vec{r}_i$ and $\vec{r}'_i$ are vectors from the centroids to the i th atom of each of the 2 conformers. When computed with respect to the α -carbon positions alone, this quantity will be given the superscript α , and when computed *versus* the

positions of all heavy atoms (i.e. non-hydrogen and non-pseudo atoms), it will be given the superscript h. Drawings of the superpositions of the structures obtained by minimizing the RMS co-ordinate difference are also used to visualize the differences between the computed structures.

Further comparisons are necessary to detect differences between conformers which are local with respect to the polypeptide chain. Thus, RMSD values were also computed between corresponding tetrapeptide segments along the chain (Havel & Wüthrich, 1985). To compare the conformations of corresponding pairs of individual amino acid residues in 2 different computed structures, RMS differences (modulo 180°) in their dihedral angles (DHAD) were evaluated. In the case of the backbone angles ϕ and ψ , the DHAD value between 2 conformers is given by:

$$\text{DHAD} = \left[\frac{1}{m} \sum_{i=1}^m \frac{1}{2} ((\phi_i - \phi'_i)^2 + (\psi_i - \psi'_i)^2) \right]^{1/2}, \quad (2)$$

where the primed and unprimed quantities refer to the 2 structures being compared.

3. Results

(a) The experimental data used as input for the distance geometry calculations

An illustration of the NOESY experiments which provided the distance constraints used to determine the spatial structure of BUSI IIA is presented in Figures 1 to 3. Figure 1 shows an expanded plot of a small region of a phase-sensitive NOESY spectrum of BUSI IIA, which was recorded with high digital resolution. Using NOESY spectra of this quality, where the resolved fine structure enables unambiguous identification of cross peaks even when they have nearly identical chemical shifts, 556 NOEs were identified. Figures 2 and 3 show data on the build-up rates of NOESY cross peaks, as they were collected from spectra recorded at lower resolution. Analysis of the complete set of NOESY spectra recorded with variable τ_m in H_2O and $^2\text{H}_2\text{O}$ using the procedures described in Materials and Methods, section (b), resulted in the distance constraints presented in Table 2. This Table was laid out so that the density of distance constraints on specified parts of the polypeptide chain becomes readily apparent. Local constraints on the backbone conformation appear in column 1, local constraints on the conformation of individual amino acid side-chains appear in column 2, medium and long-range constraints on the backbone in column 3, and long-range constraints affecting both backbone and side-chains in column 4. The previously described system of pseudo structures for amino acid residues (Wüthrich *et al.*, 1983) was used to obtain proper reference points for distance constraints to groups of protons which could not be stereospecifically assigned. An overview of the locations in the amino acid sequence of BUSI IIA which are connected by the different types of NOE distance constraints listed in Table 2 is presented by the diagonal plot of Figure 4.

The DISGEO distance geometry program used

here has been described and tested in previous publications (Havel & Wüthrich, 1984, 1985). It accepts input in the form of lower and upper bounds on the interatomic distances (distance constraints) and lower and upper bounds on the oriented volumes spanned by sets of four atoms (chirality constraints: Havel *et al.*, 1983). These constraints were obtained both from the primary structure of the protein and from the n.m.r. experiments. The n.m.r. data consisted primarily of those listed in Table 2. However, some qualitative features of the polypeptide conformation can be readily derived by inspection of Table 2 and Figure 4, and use of this information can facilitate the computations considerably.

First, a qualitative analysis of the NOESY spectra provided the information needed to establish the presence of the disulfide bridges Cys7–Cys39, Cys17–Cys36 and Cys25–Cys57. For example, there are NOEs between Ala8 NH and Cys39 $\text{C}^{\beta}\text{H}_2$, Cys17 $\text{C}^{\beta}\text{H}_2$ and Cys36 NH, and between Thr31 $\text{C}^{\gamma}\text{H}_3$ and Cys57 NH (Thr31 forms a β -bridge with Cys25: Table 2). The only NOE which could be taken to indicate the existence of an alternative disulfide bridge is between $\text{C}^{\alpha}\text{H}$ of Cys36 and NH of Cys39. However, these two residues are in a regular α -helix (Williamson *et al.*, 1984), and could not be linked by a disulfide bridge without severe distortion of the helix. The disulfide linkages 7–39, 17–36 and 25–57 are also consistent with those in numerous homologous proteins (Kato *et al.*, 1976). These disulfide bonds were fixed by imposing the distance constraints described in Materials and Methods, section (c).

Secondly, *cis* and *trans* peptide bonds can be distinguished from the sequential NOEs (Arseniev *et al.*, 1983; Wüthrich *et al.*, 1985). Using the previously established criteria, we concluded from the short sequential distance $d_{\alpha\alpha}$ between residues Asp12 and Pro13 that they are linked by a *cis* peptide bond. Similarly, the short contact from $\text{C}^{\alpha}\text{H}$ of Asn22 to $\text{C}^{\beta}\text{H}_2$ of Pro23 shows that these two residues are linked by a *trans* peptide group. Direct evidence for a *trans* conformation in all other peptide bonds comes from the observed sequential NOEs, except for the segments from residues 1 to 2, 3 to 6 and 31 to 32, where no data on the sequential connectivities was obtained (Table 2). For the preparation of the input we assumed that with the exception of Asp12–Pro13, all peptide bonds were *trans*.

The first set of structure calculations used only the NOE constraints listed in Table 2, the locations of the disulfide bonds given above, and these *cis-trans* isomers of the peptide bonds as input (i.e. the NOE data set, N). A second set of structure calculations used all the constraints of the data set N together with certain supplementary experimental constraints as input (i.e. the data set complemented with supplementary constraints, C). These supplementary constraints were derived from the locations of hydrogen bonds in the secondary structures, from the vicinal spin-spin couplings

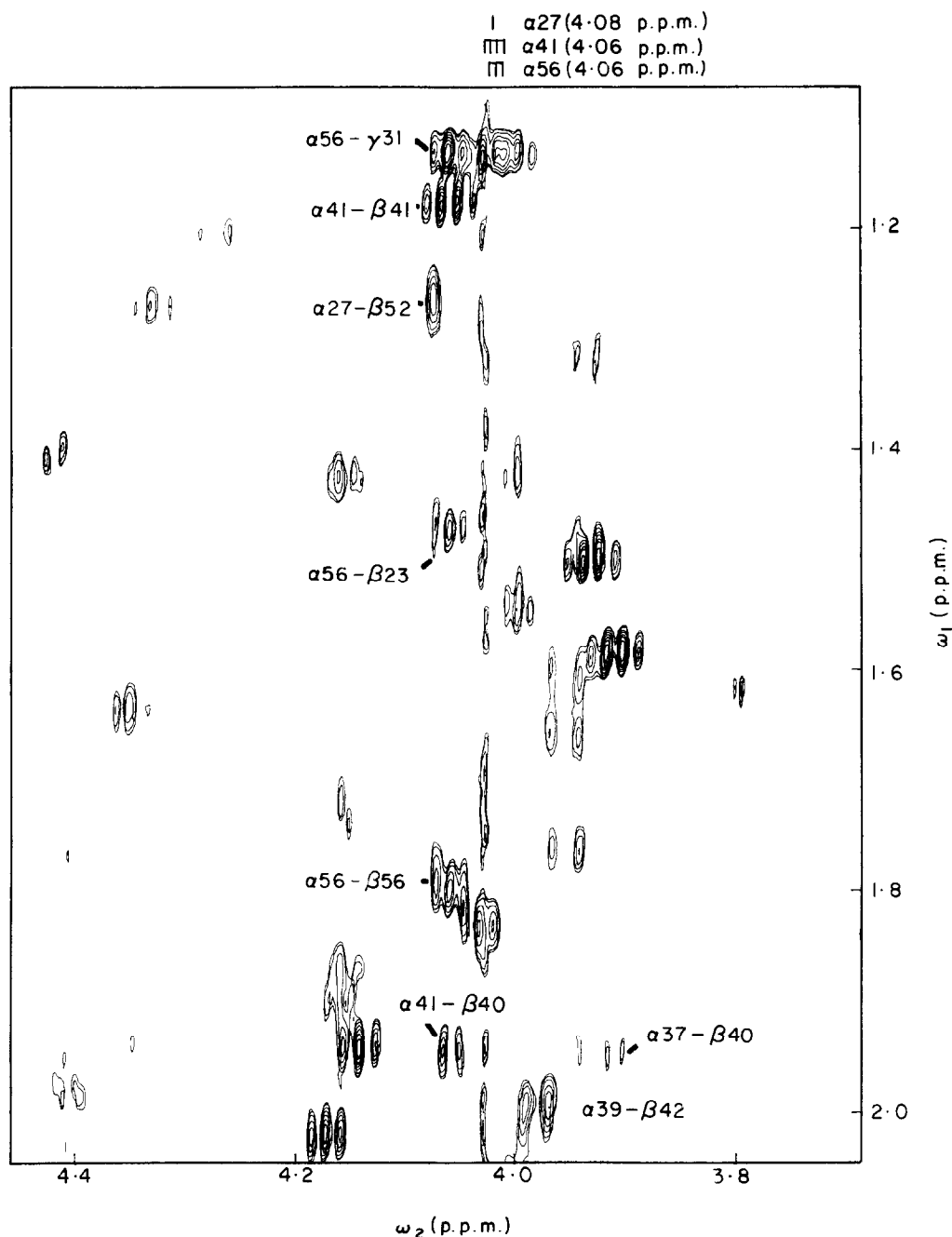


Figure 1. High-resolution absorption mode NOESY spectrum recorded at 500 MHz in a 0.016 M-solution of BUSI IIA in $^2\text{H}_2\text{O}$, $\text{p}^2\text{H} = 5.3$, $t = 45^\circ\text{C}$, $\tau_m = 150$ ms. The region ($\omega_1 = 1.1$ to 2.0 p.p.m., $\omega_2 = 3.7$ to 4.4 p.p.m.) is shown. The digital resolution after zero filling was 3.7 Hz in ω_1 and 0.45 Hz in ω_2 . Some of the cross peaks have been assigned to the pair of protons which are connected by the corresponding NOE, and the first proton listed is the one of which the chemical shift is given by ω_2 . For the C^α protons of S27, A41 and K56 the ω_2 positions and the multiplet fine structures are indicated at the top of the Figure. Although the 3 resonances fall within 0.03 p.p.m. of each other all NOEs involving these protons could be readily assigned by the spin-spin splitting observable in ω_2 . All of the peaks in this region for which the assignments have not been shown here arise from intraresidue NOEs.

$^3J_{\text{HN}\alpha}$ (Williamson *et al.*, 1984), and from $^3J_{\alpha\beta}$, as given in Figure 5. Only those spin-spin couplings which could be unambiguously interpreted in terms of the dihedral angle, and those hydrogen bonds simultaneously implied by the presence of at least two NOEs and by slow amide proton exchange rates (Wüthrich *et al.*, 1984), were used for the

supplementary constraints. Figure 5 provides a survey of the sequence locations affected by these constraints, which also include hydrogen bonds between the backbone amide protons and the side-chain carboxyl groups of Glu9 and Glu20 (Ebina & Wüthrich, 1984). The geometric conditions by which these supplementary constraints were

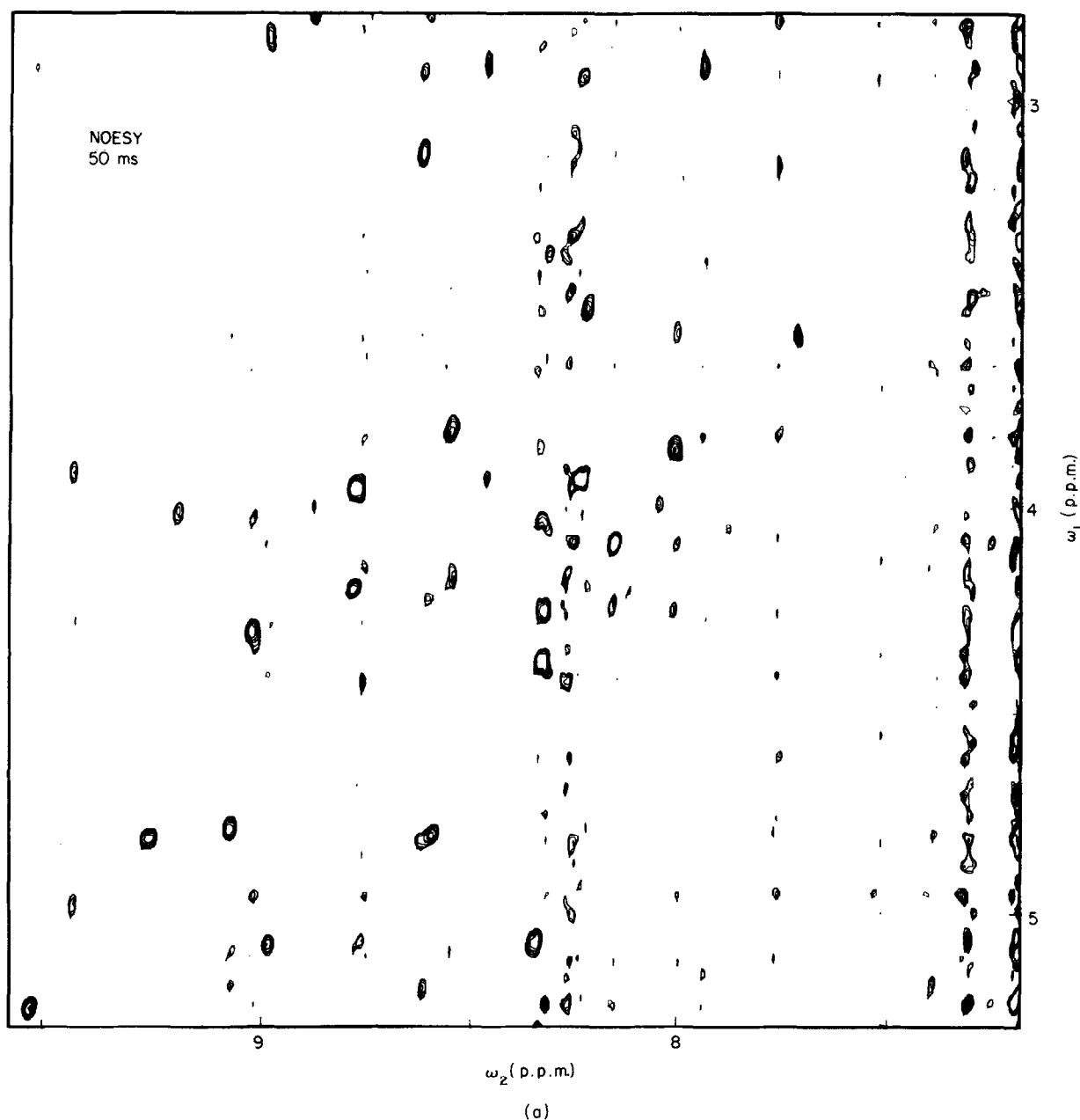


Figure 2. Low-resolution (see the text) absorption mode NOESY spectra recorded at 500 MHz in a 0.016 M-solution of BUSI IIA in H₂O, pH 5.3, $t = 45^\circ\text{C}$, with the mixing times indicated in Figures. The spectral region $\omega_1 = 2.7$ to 5.3 p.p.m., $\omega_2 = 7.2$ to 9.6 p.p.m. is shown. The digital resolution after zero filling is 4.8 Hz in both directions. Approximately 120 NOE cross peaks were resolved and assigned in this region of the spectrum. Of these, roughly 80 were intense enough and sufficiently well-resolved to be used for measurements of the τ_m dependence of the NOE intensities (see Fig. 3).

actually imposed on the computed structures have been described in Materials and Methods, section (c).

(b) *Spatial structures of BUSI IIA computed from the n.m.r. data*

In all, ten random structures† were computed using the two sets of geometric constraints as input, as described in the preceding section. The five conformers N1 to N5 computed from data set N

(above) are shown in Figure 6(a), and the five conformers C1 to C5 computed from the data set C are shown in Figure 6(b). The schematic drawing of the main features of the BUSI IIA structure (Fig. 7) may be helpful as a guide to Figure 6.

The convergence of the DISGEO program with

† These random structures were obtained by embedding random sets of distances, as described by Havel & Wüthrich (1984, 1985).

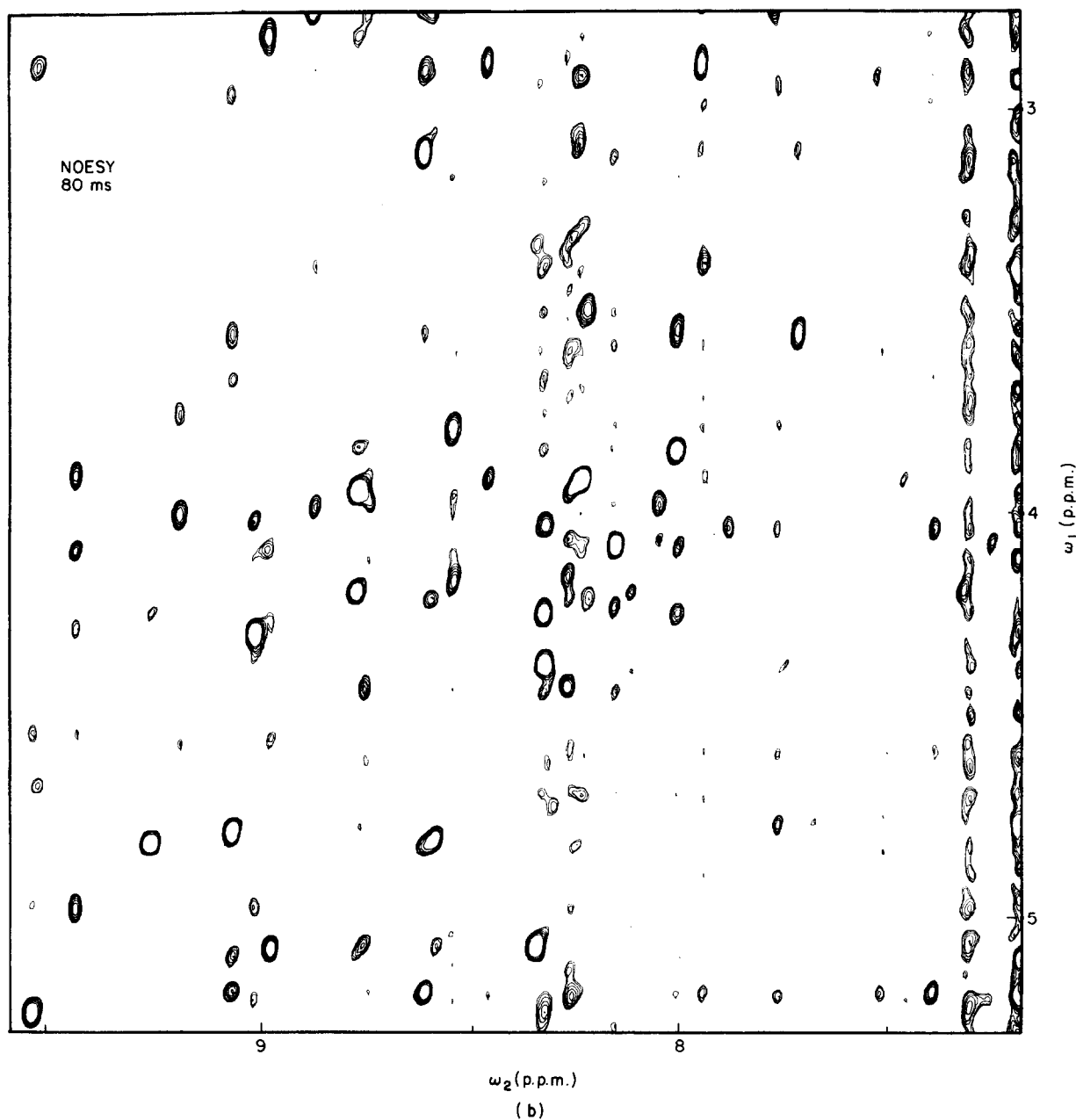
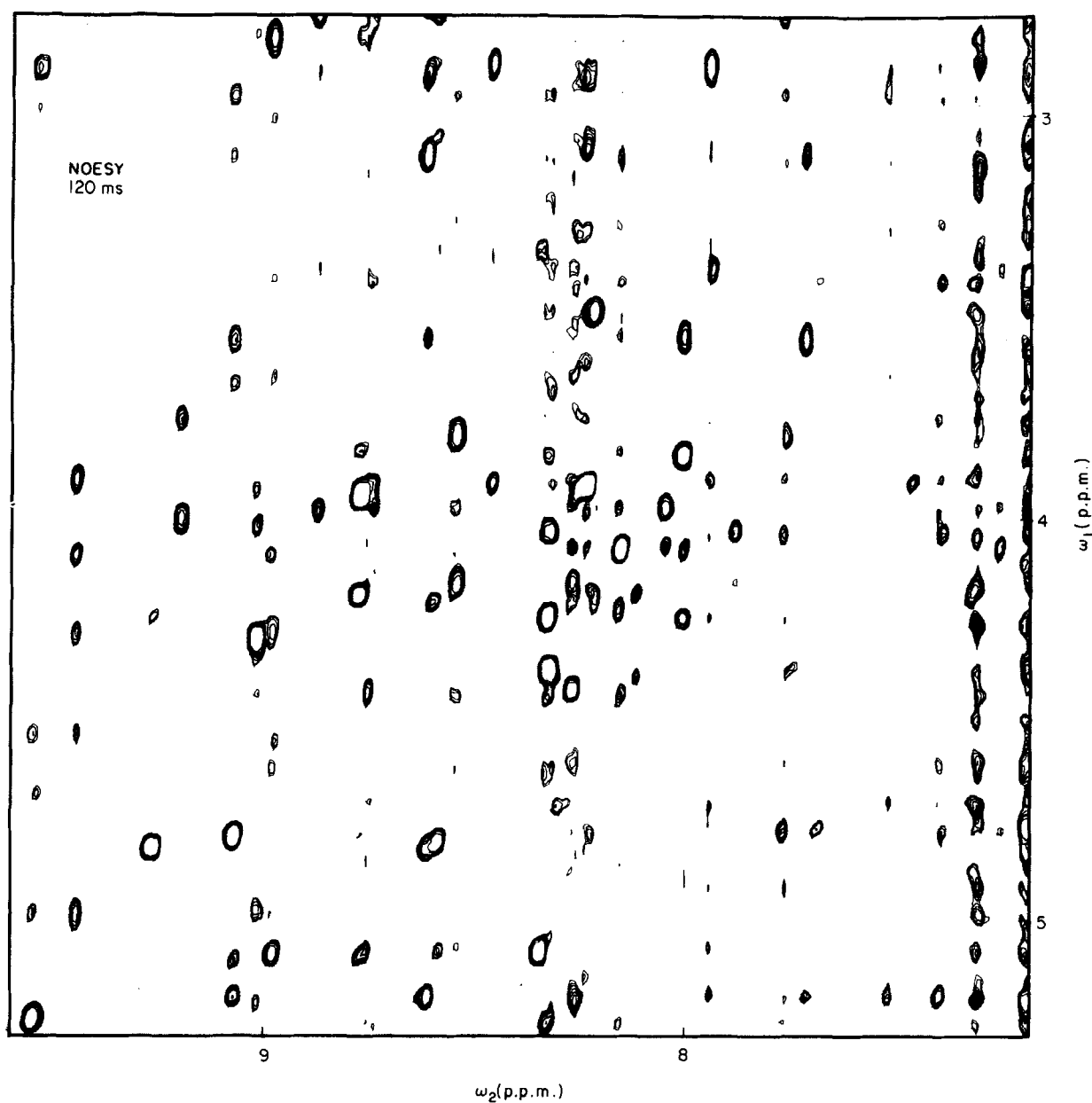


Fig. 2

both input data sets N and C was comparable to the experience gained with similar data sets which had been simulated from the crystal structure of BPTI (Havel & Wüthrich, 1985). This implies that there are no serious inconsistencies in the geometric constraints used as input. In particular, there were no significant violations of the chirality constraints in any of the ten structures. Violations of covalent bond lengths by more than 0.05 Å were rare, as were van der Waals' violations of more than 0.1 Å. Peptide bonds were always within 3° of planarity. Figure 8 shows histograms in 0.1 Å increments of the violations of all of the input distance constraints found in each of the ten structures. In the best case, C5, there were only three violations exceeding 0.2 Å, and in the worst case there were 24

such violations. No distance constraints exceeding 1.0 Å were found in any of the computed structures, and the size of the largest violation was about 0.5 Å on average.

A list of all pairs of backbone protons that were less than 5.0 Å apart in all five of the structures computed for each data set was prepared, and the NOESY spectra checked for the presence of the corresponding NOEs. Only two of these predicted NOEs were not observed, one of which involves the C α H of Thr31, which resonates at the same frequency as the water signal and was consequently bleached out by the water irradiation. The other is an NOE across the β -sheet between the C α H of Cys25 and the NH of Tyr32, which was probably too weak to be observed.



(c)

FIG. 2.

4. Discussion

The input for the present distance geometry calculations of the solution conformation of BUSI IIA consisted exclusively of geometric constraints determined by n.m.r. experiments or imposed by the primary structure of the protein. The search for spatial molecular structures compatible with these constraints started without any additional assumptions on the structures to be found, nor was any assumption made as to the existence of a single unique solution (Braun *et al.*, 1981; Havel & Wüthrich, 1984). The BUSI IIA conformers obtained from different computations with the same geometric constraints are therefore to be regarded as a random selection from the ensemble of all

conformations of BUSI IIA which are consistent with the experimental data. It may be added that this method is suitable for studies of "uniquely structured" polymers, in the sense that the atoms are confined to discrete spatial locations about which they undergo limited thermal motions (see Braun *et al.*, 1981). In systems where multiple discrete conformers would be substantially populated in a rapid exchange situation, the structures obtained with the present analysis of the NOE intensities would not necessarily correspond to conformations contributing to the true equilibrium.

Since the geometric constraints imposed by the experimental data were generally rather imprecise,

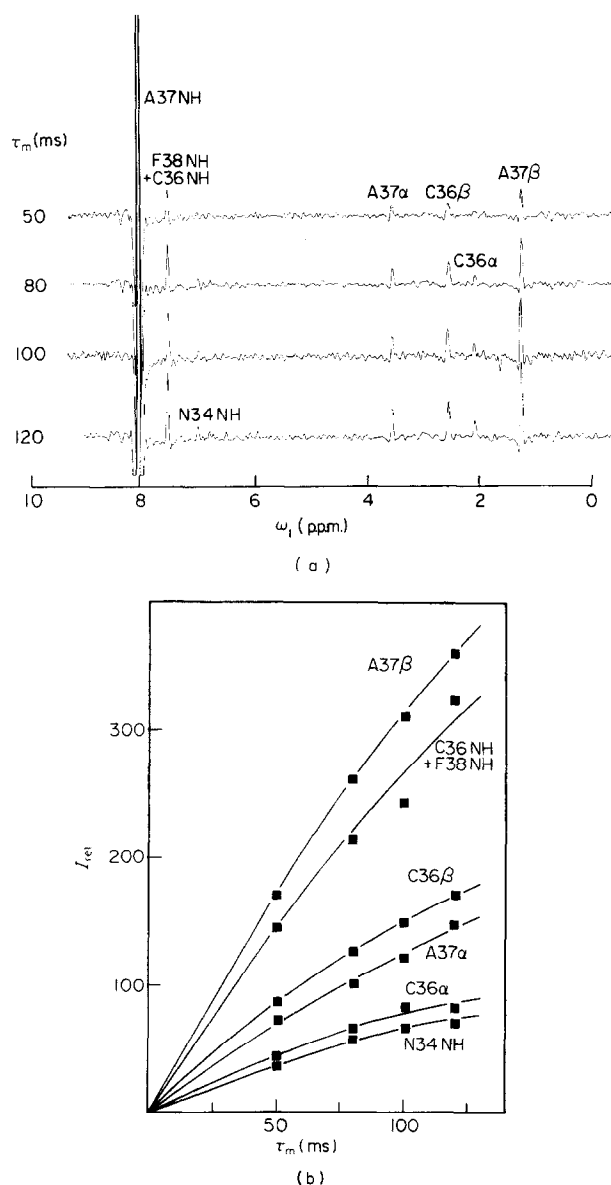


Figure 3. Build-up of NOEs with τ_m , using spectra from the same series as Fig. 2. (a) Cross-sections parallel to ω_1 at the chemical shift of the amide proton of A37 ($\omega_2 = 8.45$ p.p.m.). The peaks have been identified by the IUPAC-IUB one-letter symbol for amino acids, the position in the amino acid sequence, and the proton type involved. (b) Plot of the height of the peaks in (a) against τ_m . The vertical scale is arbitrary, since the cross peak heights were only measured relative to each other.

a certain amount of variation between the computed structures was to be expected. Nevertheless, inspection of the superpositions of the backbones of the individual conformations (Fig. 6) shows that the structures computed from each data set have many features in common. In the following, we first describe the global features of the BUSI IIA solution conformation. We then examine the high resolution crystal structures of two

homologous proteins, and quantitatively compare them with the conformers of BUSI IIA computed from the n.m.r. data. Finally, the conformers of BUSI IIA are compared with one another in order to decide what features of the conformation are uniquely determined by them.

(a) The solution conformation of BUSI IIA

The elements of regular secondary structure, shown schematically in Figure 7, include a segment of distorted 3_{10} -helix from residues 8 to 11, a quite regular α -helix from residues 34 to 45, and a triple-stranded, antiparallel β -sheet with strands extending from residues 23 to 27, from 29 to 33 and from 50 to 55, with a β -bulge formed by residues 52 and 55. The locations of these regular secondary structure elements along the amino acid sequence coincide with the results of earlier work, where the sequence locations of the helical and β -sheet structures were determined by an empirical approach relying on visual inspection of the n.m.r. data (Williamson *et al.*, 1984). It is particularly pleasing to observe this good agreement with the earlier predictions even in the structures N1 to N5, which were calculated from the NOE distance constraints alone (Table 2). Among the five structures C1 to C5 computed with the inclusion of hydrogen bond and ϕ angle constraints (Fig. 5), the α -helix and the strand of the β -sheet from residues 23 to 27 show a lesser spatial dispersion than among N1 to N5 (Fig. 6).

For both sets of conformers, N1 to N5 and C1 to C5, the molecule is roughly an oblate ellipsoid in shape, with maximum dimensions of about $25 \text{ \AA} \times 25 \text{ \AA} \times 15 \text{ \AA}$, and average radii of gyration of the C^α positions about the three principal axes equal to $11.7 \text{ \AA}$, $11.5 \text{ \AA}$ and $9.7 \text{ \AA}$. The variations in these radii of gyration between the different computed structures were less than 5%. The topology of the regular secondary structure elements and the other regions of the polypeptide chain is also very similar in all ten of the computed conformers. In the orientation of the protein chosen in Figures 6 and 7, the polypeptide segment 5–22 describes a large loop on the right. Except for the distorted helical segment near residue 10, this loop consists of irregular random coil secondary structure, and also includes a *cis* peptide bond between residues 12 and 13. The loop is held against the α -helix from residues 34 to 45 by the disulfide bonds 7–39 and 17–36, as well as by several long-range NOEs (Table 2). In contrast, there are only three NOE contacts between this loop and the β -sheet, all of which involve the side-chain of Val5 (Table 2). Figures 6 and 7 indicate that there could be additional contacts between the N-terminal residues 1 to 4 and the β -sheet. Since only a single sequential NOE was observed in this region, a likely explanation for the absence of further NOEs is that this segment is rather mobile in the solution conformation.

In the triple-stranded β -sheet, the antiparallel alignment of the three strands is unambiguously

Table 2
Distance constraints (Å) between groups of hydrogen atoms in the three-dimensional structure of BUS1 IIA obtained from NOESY spectra in H₂O and ²H₂O

Sequential				Intraresidue				Long-range backbone				Long-range			
Pyr1															
Gly2															
HN	A3	HN	4.0												
Ala3															
HN	G2	HN	4.0												
Gln4															
Val5															

Table 2 (continued)

Sequential		Intraresidue				Long-range backbone				Long-range																	
										H4	L51	QD	7.4	<i>q</i>	(6-12, 0-67/0-19)												
										PB	L51	QD	8.4	<i>m_q</i>	(2-31, 0-19)												
Cys25	HN	H24	HA	2.5	HN	C25	PB	3.6	<i>i</i>	(8-29, 2.57) (8-29, 1.85)	HA	T31	HA	4.0	(5-19, 4.65) (8-29, 7.06) (8-29, 3.83)	HN	L51	QD	7.4	<i>q</i>	(8-29, 0-19)						
	HA	G26	HN	2.5							HN	H53	HN	4.0		HN	L51	QD	8.4	<i>m_q</i>	(2-31, 0-19)						
	HN	G26	HN	2.5							HN	R54	HA	4.0													
Gly26	HN	C25	HA	2.5							HN	E30	HN	4.0	(9-53, 7.72)	PA	F38	C4	8.0	<i>m_r</i>	(4-94, 7.03)						
	HN	C25	HA	2.5							PA	L51	HA	5.0	<i>m</i> (4-94, 4.26)	PA	I49	MG	7.0	<i>m_m</i>	(4-94, 0.88)						
Ser27	HN	N28	HN	4.0	HN	S27	PB	3.6	<i>i</i>	(9-43, 4.27) (9-43, 3.89)	HA	G29	HN	4.0	(4-08, 8.52) (9-43, 1.32) (9-43, 8.74) (4-08, 1.28)	PB	F38	QR	8.4	<i>m_q</i>	(4-27, 6.87/7-03) (9-43, 0.88)						
	HN	I49	HB	5.0																							
	HN	N50	HN	4.0																							
	HA	K52	PB	6.0							<i>m</i>																
Asn28	HN	S27	HN	4.0							HN	E30	HN	4.0	(8-95, 7.72)	HN	I49	MG	6.0	<i>m</i>	(8-95, 0.88)						
	HN	G29	HN	3.0																							
Gly29	HN	N28	HN	3.0							HN	S27	HA	4.0	(8-52, 4.08)												
	HN	E30	HN	3.0																							
Glu30	HN	G29	HN	3.0	HN	E30	PB	3.6	<i>i</i>	(7-72, 1.94) (7-72, 1.65)	HN	G26	HN	4.0	(7-72, 9.53) (7-72, 8.95)												
	HA	T31	HN	3.0							HN	N28	HN	4.0													
Thr31	HN	E30	HA	3.0	HA	T31	MG	3.1	<i>i</i>	(4-65, 1.13)	HA	C25	HA	4.0	(4-65, 5.19)	MG	P23	HA	6.0	<i>m</i>	(1-13, 4.78)						
Tyr32					HN	Y32	PB	3.6	<i>i</i>	(9-50, 3.02) (9-50, 2.91)	HN	H24	HN	4.0	(9-50, 9.25)	HN	T31	MG	6.0	<i>m</i>	(9-50, 1.13)						
Gly33	HN	N34	HN	4.0							PA	P23	HA	5.0	<i>m</i> (3-75, 4.78)	PA	T31	MG	7.0	<i>m_m</i>	(3-75, 1.13)						
	HN	N34	HN	4.0							HN	A37	MB	6.0	<i>m</i> (9-18, 1.58)												
Asn34	HN	G33	HN	4.0							HA	N22	HN	4.0	(5-13, 9.05)	ND	E20	HN	6.0	<i>m</i>	(7-78, 8.07)						
	HA	K35	HN	2.5							HA	N22	PB	6.0	<i>m</i> (5-13, 2.96)	ND	E20	PB	7.0	<i>m_m</i>	(7-78, 2.02)						
	HN	K35	HN	4.0							PB	N22	HN	6.0	<i>m</i> (3-55/3.16, 9.05)	HA	H24	H2	5.0		(5-13, 7.53)						
	PB	K35	HN	3.5	<i>m</i>						PB	C36	HN	6.0	<i>m</i> (3-16, 7.92)												
											HN	A37	HN	4.0	(7-37, 8.45)												
											HN	A37	MB	6.0	<i>m</i> (7-37, 1.58)												
											HN	F38	HN	4.0	(7-37, 7.92)												
Lys35	HN	N34	HA	2.5							HN	N22	PB	6.0	<i>m</i> (8-60, 3.02)	HN	N22	ND	6.0	<i>m</i>	(8-60, 7.76)						
	HN	N34	HN	4.0							HA	F38	HN	4.0	(2-93, 7.92)	HA	H24	H2	5.0		(2-93, 7.53)						
	HN	N34	PB	3.5	<i>m</i>						HA	F38	PB	6.0	<i>m</i> (2-93, 2.78/2.72)	HN	H24	H2	5.0		(8-60, 7.53)						
	HN	C36	HN	3.0												PB	H24	H2	6.0	<i>m</i>	(1-72, 7.53)						
	PB	C36	HN	4.0	<i>m</i>																						
Cys36	HN	K35	HN	3.0							HN	C17	PB	6.0	<i>m</i> (7-92, 3.35)	HA	F10	C4	7.0	<i>r</i>	(2-44, 7.27)						
	HN	K35	PB	4.0	<i>m</i>						HN	N34	PB	6.0	<i>m</i> (7-92, 3.16)	HA	F10	C1	7.0	<i>r</i>	(2-44, 7.16)						
	HN	A37	HN	3.0							HA	C39	HN	4.0	(2-44, 8.83)	HN	F10	C4	7.0	<i>r</i>	(7-92, 7.27)						
	PB	A37	HN	4.0	<i>m</i>											PB	F10	C4	8.0	<i>m_r</i>	(2-93, 7.27)						
Ala37	HN	C36	HN	3.0							HN	C17	PB	6.0	<i>m</i> (8-45, 3.35)	MB	Y32	PB	7.0	<i>m_m</i>	(1-58, 3.02)						
	HN	C36	PB	4.0	<i>m</i>						MB	G33	HN	6.0	<i>m</i> (1-58, 9.18)	MB	Y32	QR	8.4	<i>m_q</i>	(1-58, 7.10)						
	HN	F38	HN	3.0							HN	N34	HN	4.0	(8-45, 7.37)												
	MB	F38	HN	4.0	<i>m</i>						MB	N34	HN	6.0	<i>m</i> (1-58, 7.37)												
											HA	K40	HN	4.0	(3-91, 8.72)												
											HA	K40	PB	6.0	<i>m</i> (3-91, 1.97)												
Phe38	HN	A37	HN	3.0	HN	F38	PB	3.1	<i>i</i>	(7-92, 2.78) (7-92, 2.72)	HN	N34	HN	4.0	(7-92, 7.37) (7-92, 2.93) (2-78/2.72, 9.93) (3-40, 7.23) (3-40, 1.16)	QR	V5	QG	9.8	<i>q_q</i>	(6-87, 1.01)						
	HN	A37	MB	4.0							HN	K35	HA	4.0		QR	H24	HA	7.4	<i>q</i>	(6-87, 6.12)						
	HN	C39	HN	3.0							PB	K35	HA	6.0		C4	G26	PA	8.0	<i>r_m</i>	(7-03, 4.94)						
											HA	A41	HN	4.0		QR	S27	PB	8.4	<i>q</i>	(6-87/7.03, 4.27)						
											HA	A41	MB	6.0		HA	Y32	QR	7.4	<i>q</i>	(3-40, 7.10)						
Cys39	HN	F38	HN	3.0							QR	Y32	PB	8.4	<i>q_m</i>	(6-87, 3.02)											
	HN	K40	HN	3.0							QR	C39	HN	7.4	<i>q</i>	(6-87, 8.83)											
											QR	V42	QG	9.8	<i>q_q</i>	(6-87/7.03, 0.10)											
											QR	I49	MG	8.4	<i>q_m</i>	(6-87/7.03, 0.88)											
											QR	N50	HA	7.4	<i>q</i>	(6-87, 5.03)											
											QR	L51	HN	7.4	<i>q</i>	(6-87/7.03, 8.95)											
											QR	L51	QD	9.8	<i>q_q</i>	(6-87/7.03, 0.19)											
Lys40	HN	C39	HN	3.0	HN	K40	PB	3.6	<i>i</i>	(8.72, 1.97)	HN	A37	HA	4.0	(8.72, 3.91)												

Table 2 (continued)

[illegible]

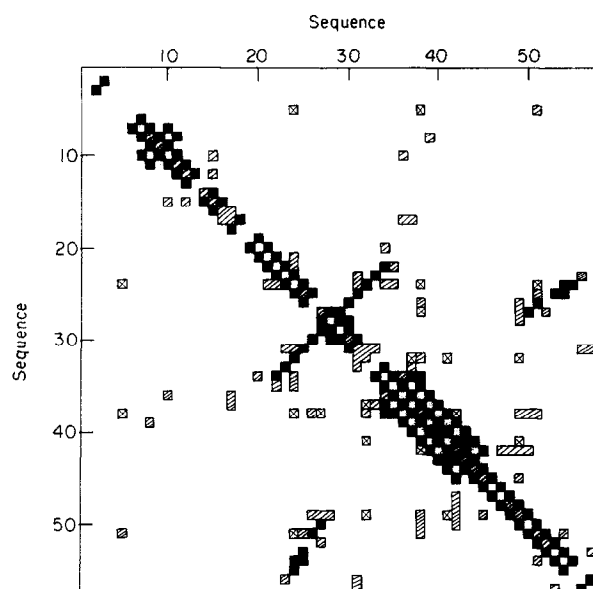


Figure 4. Diagonal plot of the NOEs used for the calculation of the tertiary structure of BUSI IIA. Both axes are calibrated with the sequence of the protein. A filled square at the position (x, y) in the plane indicates that a NOE between backbone protons (NH or $\text{C}^{\alpha}\text{H}$) of the 2 residues in the sequence locations x and y was observed. A hatched square indicates a NOE between the backbone amide or protons of one residue and any of the side-chain protons in another residue, while a square containing a cross indicates a NOE between protons in the side-chains of 2 different residues. Where 2 residues are connected by more than 1 NOE, only the one which involves the greater number of backbone protons is shown. Further details on the NOEs may be found in Table 2.

determined by numerous interstrand NOEs (Table 2; Williamson *et al.*, 1984). The sheet has a pronounced, right-handed twist in all ten of the

computed structures. In Figures 6 and 7, the sheet lies to the left of the α -helix 34–45, and numerous NOEs between residues located in the α -helix and β -sheet show that these two principal secondary structure elements are tightly packed together in the tertiary structure. The C-terminal residue Cys57 is tied to the β -sheet by the disulfide bond with Cys25. The observation of numerous NOEs (Table 2) provided direct evidence that the C-terminal region of the polypeptide chain has a unique spatial arrangement in the solution conformation. The architecture of the β -sheet region further includes two tight turns, one from residues 27 to 29, where the first two strands form a “hairpin”, and the other from residues 47 to 49.

Overall, the topology of the solution conformation of BUSI IIA is thus characterized by the α -helix 34 to 45 being sandwiched between the extensive loop from residues 5 to about 20 and the β -sheet. In this spatial arrangement, the α -helix forms the core of the molecule, which provides an explanation for the outstandingly slow exchange of the hydrogen-bonded amide protons in the helix (Wüthrich *et al.*, 1984).

(b) Comparison with the crystal structures of homologous proteins

Although no crystal structure exists for BUSI IIA, high resolution crystal structures do exist for three homologous proteins, e.g. the porcine pancreatic secretory trypsin inhibitor (PSTI) (Bolognesi *et al.*, 1982), the third domain of Japanese quail ovomucoid (OMJPQ3) (Papamokos *et al.*, 1982), and the third domain of turkey ovomucoid (Fujinaga *et al.*, 1982). Since the two ovomucoids are very similar in sequence and tertiary structure, we have chosen to compare BUSI IIA with PSTI and OMJPQ3. OMJPQ3 has

The amino acid residues are identified either by the 3-letter symbol or the 1-letter symbol followed by a number indicating the position in the amino acid sequence. HN stands for amide proton, HA for C^{α} proton, HB for C^{β} proton, HG for C^{γ} proton and HD for C^{δ} proton. H2 and H4 are the protons attached to the ring carbons 2 and 4 of His. The following upper case letters indicate pseudo atoms which substitute for a group of 2 or more protons, and lower case letters indicate that the distance constraint indicated has been modified to account for the use of the pseudo atom (Wüthrich *et al.*, 1983). M replaces the 3 protons of a methyl group, and is located in the center of the 3 proton positions, with the correction factors $m = 1.0 \text{ \AA}$ for long-range constraints and $i = 0.6 \text{ \AA}$ for certain intraresidue constraints. P replaces the protons of a methylene group and is located in the center between the 2 proton positions, with the same corrections m and i as for methyl groups. Q replaces the 2 methyl groups of Val and Leu and is located centrally with respect to the 2 methyl groups, with corrections of $q = 2.4 \text{ \AA}$, or $s = 1.7 \text{ \AA}$, respectively, for long-range constraints or for certain intraresidue constraints. QR is at the center of the ring protons C2H, C3H, C5H and C6H in Phe or Tyr, also with a correction of $q = 2.4 \text{ \AA}$. Where the C2H and C6H resonances could be resolved from those of C3H and C5H, distances to the ring protons were referred to the ring carbon atoms C-1 and C-4, respectively, with a correction of $r = 2.0 \text{ \AA}$. Distance constraints to the amide protons in the side-chains of Asn22 and Asn34 are referred to the amide nitrogen atom ND, with a correction of $m = 1.0 \text{ \AA}$.

The rows following the 3-letter symbol for the amino acid list all the distance constraints that involve hydrogen atoms of this residue. For each residue the distance constraints are grouped into 4 classes (see the text), which are presented in the 4 columns. Sequential constraints are those between backbone protons or between a backbone and a C^{β} proton in sequentially neighboring residues. Intraresidue constraints are between side-chain hydrogen atoms and any other proton of the same residue. Long-range backbone constraints are between 2 backbone C^{α} or amide protons, or between a backbone proton and the C^{β} protons in residues that are not nearest-neighbors in the sequence. Long-range constraints are all those that have not been listed elsewhere. With the exception of the intraresidue constraints, all entries appear twice, once for each residue concerned. In each column the first entry identifies a hydrogen atom in the amino acid that is identified by the 3-letter code. The second and third entries indicate a different residue and the hydrogen atom (or pseudo atom) in this residue to which a distance constraint has been observed. The fourth entry is the distance constraint in $\text{\AA}$, possibly with symbols indicating that 1 or 2 correction factors were added to the NOESY distance constraints to allow for the use of pseudo atoms. The fifth entry indicates the location of the NOESY cross peak in the ω_1 – ω_2 plane in p.p.m. (Figs 1 and 2). (This fifth entry has been omitted for the sequential constraints, since these were previously described in detail (Štrop *et al.*, 1983).) Where two p.p.m. values are given separated by an oblique stroke, 2 resonance lines were resolved for the group of protons indicated.

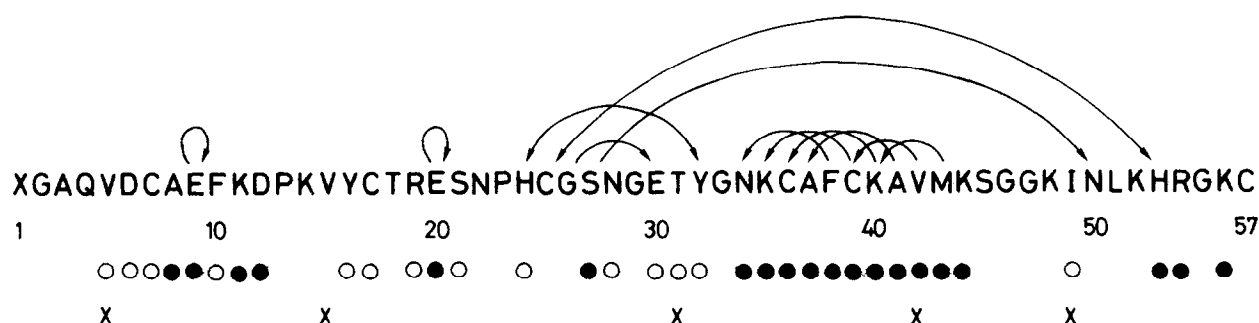


Figure 5. Diagram showing the sequence locations of the hydrogen bonds (arrows), ϕ angle constraints (circles) and χ^1 angle constraints (crosses) used as supplementary constraints in the computations of the structures C1 to C5 (see the text). The sequence of BUSI IIA is given in the standard one-letter amino acid code. The arrows point from the amide proton to the carbonyl oxygen of each hydrogen bond (2-headed arrows identify an antiparallel β -bridge). A filled circle ($\bullet$) indicates a $^3J_{\text{HN}\alpha} \leq 5.5$ Hz, so that the ϕ angle was constrained between -90° and -40° , and an open circle ($\circ$) indicates a $^3J_{\text{HN}\alpha} \geq 8.0$ Hz, so that the ϕ angle was constrained between -160° and -80° . (The present Figure differs from Fig. 7 of Williamson *et al.* (1984) in that residues K14, V15, T18, C25, L51 and K52, which have $^3J_{\text{HN}\alpha} = 7.5$ Hz, do not carry open circles because of the different cutoff applied; the 3J value of 5.5 Hz for C57 was not exploited because the ϕ angle could not be constrained properly in the amino acid residue set used.) The crosses indicate that the $^3J_{\alpha\beta}$ values were ≥ 8.0 Hz (i.e. V5, $^3J_{\alpha\beta} = 8.0$ Hz; V15, $^3J_{\alpha\beta} = 9.0$ Hz; T31, $^3J_{\alpha\beta} = 10.0$ Hz; V42, $^3J_{\alpha\beta} = 10.0$ Hz; I49, $^3J_{\alpha\beta} = 10.0$ Hz), and that the χ^1 angle was therefore constrained so as to keep the H^α and H^β protons within $\pm 30^\circ$ of the *trans* orientation.

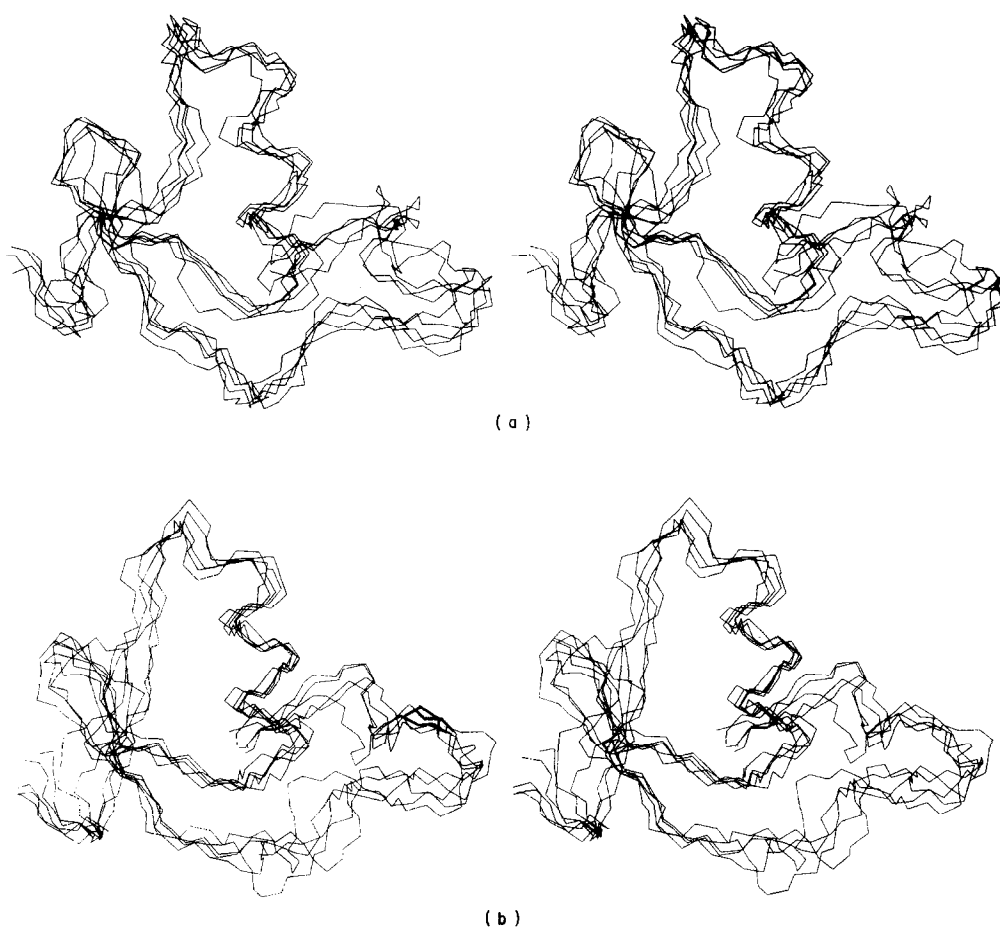


Figure 6. (a) Stereo view of the 5 structures N1 to N5 calculated using the NOE distance constraints alone (Table 2). The structures were superimposed so as to minimize the RMSD with respect to the structure N1. All main-chain backbone atoms (C α , C', N) are shown. For clarity of presentation the residues 1 to 4, which are essentially unconstrained by the n.m.r. data (Table 2), have been omitted, and the disulfide bonds are not shown. (b) Stereo view of the 5 structures calculated from the NOE constraints used in (a) plus the supplementary constraints shown in Fig. 5. As in (a), all main-chain backbone atoms are shown, the first 4 residues and the disulfide bonds have been omitted, and the 5 structures have been superimposed for best fit with the structure C1.

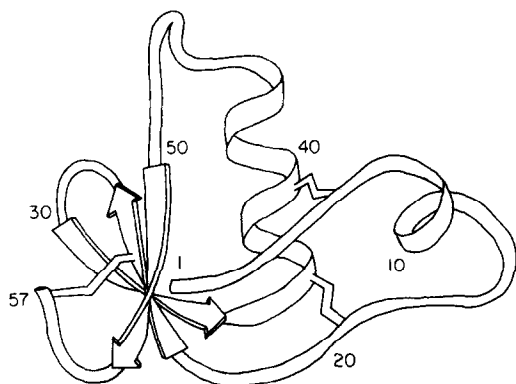


Figure 7. Schematic drawing of the polypeptide backbone "topology" which was obtained in all 10 of the BUSI IIA structures computed from the n.m.r. data presented.

45% homology and a deletion of two residues with respect to BUSI IIA. PSTI has 26% homology and a deletion of three residues. The deletions are both in the N-terminal part of the molecules, which is generally of variable length in this class of secretory proteinase inhibitors (Kato *et al.*, 1976). We have therefore used only the C-terminal part of the molecules (from residues 23 to 57 in BUSI IIA and 22 to 56 in the two homologues) for the purpose of comparison.

The average RMSD values between the α -carbons of these segments in PSTI and the BUSI

IIA structures are 2.20 Å and 2.24 Å for the N and C data sets, respectively. For OMJPQ3 the corresponding averages are 1.89 Å and 1.75 Å. The RMSD between the α -carbons in the segments 22 to 56 of PSTI and OMJPQ3 is 1.34 Å. These RMSD values indicate a high degree of similarity between the computed BUSI IIA structures and the crystal structures of these homologous proteins. This similarity is also evident in the superpositions of this region of the backbone of the BUSI IIA structure C5 on the corresponding residues of the homologues (Fig. 9), which shows that the α -helix and the β -sheet have both identical sequence locations and the same relative spatial orientations in the three proteins. Figure 9 shows further that the conformations of the N-terminal segments (residues 1 to 22) are markedly different in the three proteins. Since there are deletions of two and three residues, respectively, between positions 7 and 20 in OMJPQ3 and PSTI, and since furthermore the N-terminal loop is tied to the rest of the molecule by the two disulfide bonds 7-39 and 17-36, these differences between the computed BUSI conformations and the conformations of the homologous proteins are not unexpected and are undoubtedly significant. The segment of residues 1 to 5 is not constrained by the available n.m.r. data, and in the crystal structure of OMJPQ3 it is known to be involved in close intermolecular contacts (Papamokos *et al.*, 1982).

(c) Quantitative comparison of the computed structures

All ten of the structures computed for BUSI IIA have nearly identical overall dimensions and contain the same global chain fold as described in Results, section (a). No single structure is markedly different from the others computed from the same data set. The average values of the RMSD between the C^α positions of pairs of structures (Table 3) are 1.93 Å and 2.13 Å for the data sets N and C, respectively. These are slightly larger than those obtained with comparable simulated data sets (e.g. those labeled as III, IV, IX and X) in the previous model studies with BPTI (Havel & Wüthrich, 1985), primarily because of the uneven distribution of NOE distance constraints obtained with BUSI IIA, which are concentrated in the C-terminal half of the molecule (see Table 2 and Fig. 5). When the RMSD values were calculated between the α -carbons of residues 23 to 57 alone, they averaged 1.59 Å and 1.84 Å for the data sets N and C, respectively. The average RMSD values between all heavy-atoms in the BUSI IIA structures (Table 3) were 2.98 Å and 3.34 Å for the data sets N and C (2.55 Å and 2.84 Å for the residues 23 to 57 alone), which are also of the same order of magnitude as those obtained in the above-mentioned studies with BPTI. They indicate that the location of a large proportion of the side-chains is similar in the different computed structures. This is probably largely a consequence of the stringent packing requirements in the interior of the protein.

N1	1517	47	5	5	2	1			
N2	1564	43	9	3	4	1	0	1	
N3	1502	66	16	2	0	2	1	1	2
N4	1510	57	12	2	6	0	0	1	
N5	1499	48	4	6	3	1			
C1	1772	33	8	5	3				
C2	1658	27	6	2	0	1			
C3	1660	38	10	5	0	3	1	0	2
C4	1706	26	6	2	5				
C5	1588	14	2	0	1				

0.1 0.3 0.5 0.7
Å

Figure 8. Histograms presenting the residual violations of the distance constraints found in each of the 10 structures of BUSI IIA computed from the n.m.r. data presented. The 5 structures which were obtained using the NOE constraints alone (Table 2) are labeled as N1 to N5, while the 5 structures computed using the NOE distance constraints together with the supplementary constraints of Fig. 5 are labeled as C1 to C5. In each histogram, the number of distance constraint violations which fall within each 0.1 Å interval is given. The total number of distance constraint violations is about 40 to 60% of the total number of input distance constraints.

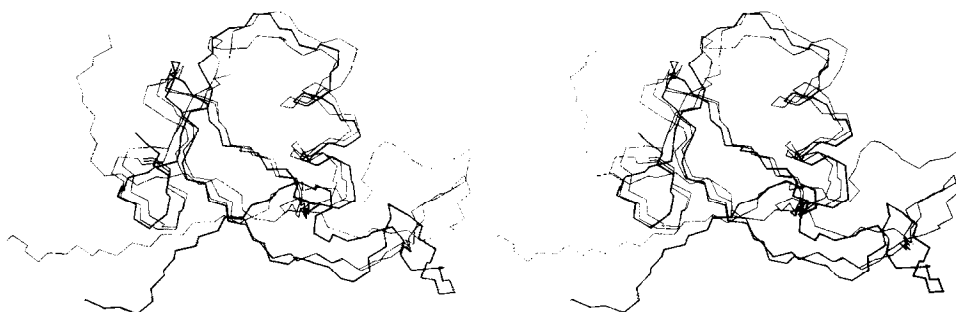


Figure 9. A comparison of the structure of BUSI IIA (heavy line) obtained from the n.m.r. data shown in Table 2 and Fig. 5 with the crystal structures of 2 homologous proteins: the porcine pancreatic secretory inhibitor and the third domain of the Japanese quail ovomucoid. The BUSI IIA structure C5 (Fig. 8) was used for this comparison. Only the backbone atoms and bonds connecting them are shown. The structures have been aligned for a minimum RMS co-ordinate difference between the α -carbons of residues 23 to 57 in BUSI IIA and the homologous residues 22 to 56 of the other 2 proteins.

It may be seen (Table 3) that the RMSD values between structures computed from the different data sets N and C are usually somewhat larger than between those computed from the same constraint set. A rather surprising result is that even though the C data set included all of the constraints of the N data set plus additional supplementary constraints (see Results, section (a)), the RMSD values between pairs of structures computed from the C data set are slightly larger in general than those between pairs computed from the N data set. Most of this increase is due to increased variability in the first strand of the β -sheet and the adjoining residues (numbers 15 through 30; Fig. 6). It can be explained by large-scale changes in the position of this segment relative to the rest of the molecule which are concomitant with changes in the twist of the β -sheet. Such "concerted motions" in the β -sheet were less pronounced in the structures computed from the N data set, presumably because the strands of the sheet were not tied together as tightly without the addition of hydrogen bonds. There is ample space for such "motions" in the β -sheet of BUSI IIA, since it is located near the surface of the protein and, when compared for example to the β -sheet of BPTI, tied down by a lesser number of long-range NOEs to other parts of the molecule.

Like the experimental constraints available for BUSI IIA, the aforementioned data sets III, IV, IX and X used with BPTI (Havel & Wüthrich, 1985) consisted of constraints which were both incomplete and imprecise, but still allowed the global features of the crystal structure from which they had been extracted to be faithfully reconstructed. The experience gained in these model studies further showed that such data are not sufficient to define precisely the local structure of the polypeptide chain, which would require a dense network of very precise local constraints. The data used here do not have the necessary precision because of the current limitations on our ability to quantitate the distance measurements obtained by NOESY, and a lack of stereospecific assignments for the methylene protons. In accord with these expectations, the measures of the differences between the local conformations of the computed structures were relatively large. The RMSD values between the α -carbons of corresponding pairs of tetrapeptide segments in the different structures averaged 0.73 Å and 0.69 Å for the data sets N and C, respectively. The DHAD values between the ϕ and ψ angles (see Materials and Methods, section (e)) averaged 77° and 51°. An exception to this general observation is the α -helix (residues 34 to 45), for which the local conformation was

Table 3
RMS deviations between all pairs formed from the ten structures computed for BUSI IIA

N1	2.04	1.80	1.76	1.57	2.45	2.29	2.57	2.33	2.10
3.08	N2	2.17	2.15	2.12	2.38	2.37	2.38	2.20	1.94
2.71	2.94	N3	1.94	1.78	2.50	2.53	2.60	2.43	2.22
2.79	2.93	2.71	N4	2.03	2.33	2.45	2.92	2.31	2.18
2.39	3.13	2.73	2.83	N5	2.52	2.29	2.50	2.30	2.12
3.90	3.95	4.08	3.67	3.89	C1	2.25	2.44	1.99	2.00
3.45	3.38	3.57	3.20	3.37	3.35	C2	2.20	2.28	2.04
3.71	3.74	3.84	3.96	3.57	3.22	3.12	C3	2.05	2.32
3.44	3.43	3.72	3.61	3.37	3.04	3.38	2.73	C4	1.71
3.09	2.96	3.25	3.19	3.20	3.26	3.05	3.27	2.96	C5

The above diagonal entries are between α -carbons alone, the below diagonal entries are between all heavy atoms. The symbols on the diagonal are those used in Fig. 8.

relatively well-defined by the n.m.r. data, as shown by average RMSD values between sets of four consecutive C^α atoms under 0.3 Å even in data set N.

Very few NOEs were observed to side-chains on the surface of the protein (Table 2). A likely explanation is that most of these side-chains NOEs were quenched by pronounced segmental mobility (Nagayama & Wüthrich, 1981). Those residues for which the χ^1 angle had been fixed by use of chirality constraints in the C data set (see Fig. 5) had relatively well-defined side-chain conformations in the computed structures, as did His24 and Phe38, both of which are buried in the interior and participate in numerous long-range NOEs (see Table 2). For His24, this fits well with the earlier observation (Štřop & Wüthrich, 1983) that this residue is unchanged over the pH range from 4.0 to about 13, and is spatially close to a titrating group which must be either a lysine or tyrosine side-chain. In the computed structures, His24 is invariably close to both Tyr32 and Lys35. Since it was previously found that Tyr32 does not titrate between pH 4 and pH 12 (Štřop & Wüthrich, 1983), this group can now be identified as Lys35. Many other side-chains in the interior were significantly restrained in their position and orientation (see the Appendix).

The investigations in this paper show that the information available from n.m.r. spectroscopy is sufficient to define the overall tertiary structure of small proteins in solution, and that algorithms based on distance geometry are a suitable means of determining the global conformation from these data. The close coincidence between the different structures computed for BUSI IIA, and between these structures and the crystal structures of two homologous proteins, is particularly gratifying when one considers that the experimental observations were used with a very conservative structural interpretation, so that the corresponding geometric constraints were quite imprecise. Since a large proportion of the available measurements concern the relative positions of protons that are near to one another in the amino acid sequence, it is perhaps surprising at a first glance to find that the local structure of both the polypeptide backbone and the side-chains is less well-defined than the global conformation. The relatively high precision with which we can determine the global conformation from imprecise distance information is a consequence of the redundancy of the long-range distance information. The local structure depends more directly on a relatively small number of short-range distances, and it can be changed substantially with small changes in these distances. As a consequence, the short-range distances would have to be measured with relatively high precision in order to determine the local conformation reasonably exactly. Alternatively, precise geometric constraints on the local structure could also be obtained from other n.m.r. experiments (Wüthrich, 1976). Because distance geometry calculations do

not take account of vicinal van der Waals' repulsions (Materials and Methods, section (d)), substantial improvement in the local conformation could also be expected to follow from subsequent energy minimization. This is further explored in the Appendix.

Appendix

Here we describe a pilot study of the use of computer graphics (Langridge *et al.*, 1982) and restrained energy minimization (Fitzwater & Scheraga, 1982; Levitt, 1983) as a means of improving the local structure (ϕ , ψ angles and side-chain conformations) of the conformations computed with DISGEO from the experimental n.m.r. data. Since the DISGEO program explicitly ignores energetic considerations while attempting to obtain a good fit to the geometric constraints, it was also of interest to demonstrate that the structures are nevertheless close to good energy minima in which all of the experimental constraints are satisfied. We selected the best convergent conformation, C5 (Table 3, main text), for the present trial refinement.

For the energy function, we used the ECEPP/2 parameters (Némethy *et al.*, 1983) with a dielectric constant of four, and left all acidic and basic groups on the side-chains unionized. Considering the preliminary nature of this work, we simplified programming by not including the internal energies of the proline residues (all prolines have the down conformation), neglecting the torsional potential about the beta-to-gamma bond to cystine, and leaving out the N-terminal amide hydrogen and the C-terminal hydroxyl group. These omissions will have a minor effect on the energies we report, but will not significantly alter the location of the energy minima. The minimizations themselves were performed with a standard conjugate gradient algorithm (Luenberger, 1973), using the dihedral angles about all rotatable bonds (including the omega angles) as the variables.

The refinement procedure consisted of six steps. First, the missing hydrogen atoms were added to the C5 structure by a method similar to that described by Billeter *et al.* (1982). The resultant structure had a high energy due to the presence of isolated van der Waals' overlaps involving some of the added protons, unfavorable dihedral interactions, and a general lack of accommodation of internal polar and charged groups in hydrogen bonds and salt bridges. In the second step of the refinement procedure, we manually adjusted the dihedral angles of long side-chains on the exterior of the molecule so as to eliminate energetically unfavorable eclipsed conformations and isolated contacts with adjacent sections of the backbone. This was done with a special computer graphics program, CONFOR, which allowed us to simultaneously monitor the NOE constraints and so avoid making changes which would violate them (M. Billeter, M. Engeli & K. Wüthrich, unpublished

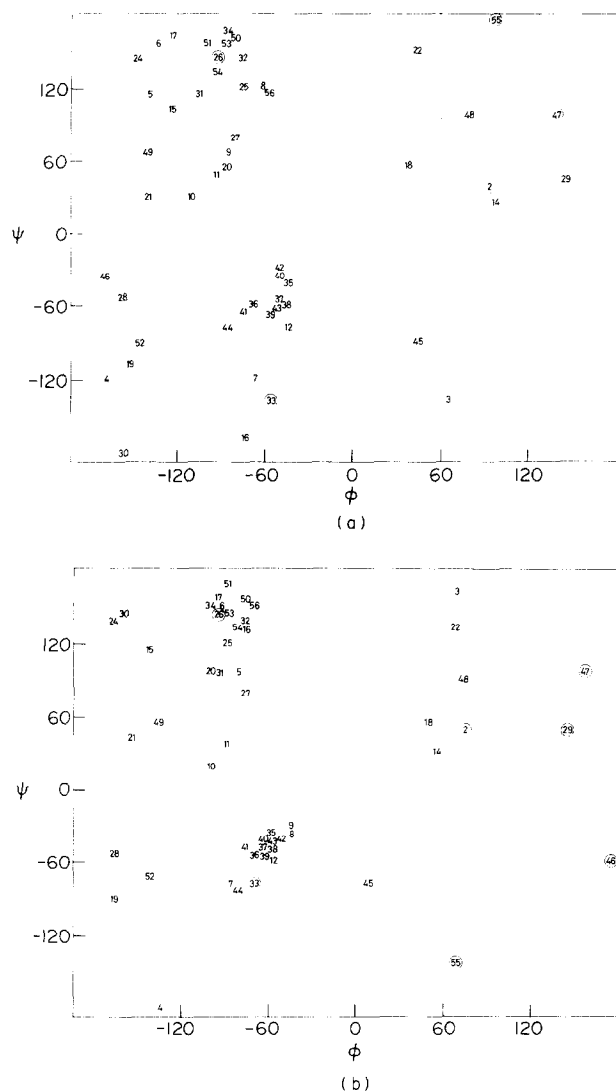


Figure A1. The Ramachandran map of the C5 pseudo structure of BUSI IIA (a) before and (b) after the refinement procedure. The sequence number of each residue in the conformation is printed at the position of its ϕ (horizontal) and ψ (vertical) angles on the axes. The sequence numbers of the glycyI residues have been circled.

work). Further optimization of the van der Waals' packing and perfection of the covalent geometry was achieved by the third step, in which a numerical optimization of the resulting structure was performed with respect to the same error function as used by the DISGEO program (Havel *et al.*, 1983), with the Cartesian co-ordinates of the atoms as the variables. The only constraints now used in calculating the error function, however, were the ECEPP covalent geometries (Némethy *et al.*, 1983) and the full van der Waals' radii of all of the atoms. The n.m.r. constraints were not included in the calculation, except for the hydrogen bonds indicated in Figure 5 of the main text. In our experience, structures subjected to the pre-processing procedure of steps two and three attain substantially lower energies on subsequent minimization.

During the optimization of covalent geometry and van der Waals' packing, approximately 10% of the NOE constraints became violated by between 1 and 3 Å. The only large dihedral angle violations were in the Glu20 and Asn34 ϕ angles, which were off by about 50°. In order to restore consistency with the geometric constraints and at the same time adequately restrain the structure during the initial steps of the energy minimization, the error function (Havel *et al.*, 1983) was calculated from the

Table A1
Minimum RMS co-ordinate differences between the C^α atoms of homologous residues in BUSI IIA, OMJPQ3 and PSTI before and after energy minimization of the BUSI IIA structure

BUSI IIA structure C5	OMJPQ3		PSTI	
	1-56	22-56	1-56	22-56
From DISGEO	4.22 Å	1.67 Å	5.16 Å	1.44 Å
Energy minimized	3.99 Å	1.52 Å	4.62 Å	1.23 Å

The RMS difference between the C^α atoms of OMJPQ3 and PSTI is 3.88 Å for residues 1 to 56, and 1.34 Å for residues 22 to 56.

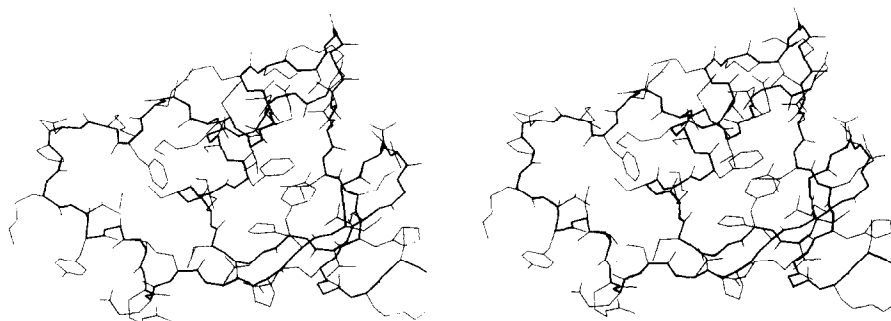


Figure A2. Stereo pair of all heavy (i.e. non-hydrogen) atoms in the energy-minimized structure of BUSI IIA obtained by refinement of the C5 pseudo structure. The bonds connecting the backbone atoms have been drawn with a heavier line. The orientation of the structure has been chosen to allow a good view of the side-chains, and is not the same as that used in Figs 6 and 7 of the main text. The disulfide bonds are not shown.

experimental distance constraints given in Table 2 and Figure 5 of the main text, and added to the energy as a pseudo potential. To do this, the interproton constraints between methylene and methyl groups were referred with the necessary corrections to the carbons of these groups. The ϕ angles indicated in Figure 5 (main text) were also restrained by means of a single-fold pseudo torsional potential which was zero between the corresponding angle bounds, and which was given a barrier height of 400 kcal (1 kcal = 4.184 kJ). The sum of the two pseudo potentials was given an initial weight of 200, and this weight decreased linearly to zero as the minimization progressed for 200 iterations, so as to attain a physically meaningful energy minimum at the end.

After this fourth step, there was only one violation of the NOE constraints exceeding 1.0 Å, and relatively few violations within the α -helix and β -sheet. In the fifth step, the remaining 1.1 Å violation between the amide proton of Ala8 and the β -carbon of Cys39 was alleviated by interactively rotating the Cys7 to Cys39 disulfide bond to the other side of its 10 kcal torsional barrier. Other, smaller NOE violations were also relieved by interactive manipulation of the Thr31, Ser45 and His53 side-chains. In the sixth and final step, the

resultant structure was subjected to another 100 cycles of restrained energy minimization with an initial pseudo potential weight of 100. The Ala8 to Cys39 violation was completely eliminated during this second minimization, but a new violation of similar magnitude appeared between the β -carbon of Val5 and the δ -ring proton of His24. Attempts at further improvement were abandoned, and the structure was accepted with 11 NOE violations exceeding 0.5 Å, and an energy of -166 kcal.

The RMS between the α -carbons of the C5 structure from DISGEO and the final energy-minimized structure was 1.69 Å. This shows that the global conformation was not changed greatly by the refinement procedure. The RMS change in the ϕ, ψ angles on refinement was 19°. Although these dihedral angle changes were not large enough to completely correct the local structure of the polypeptide, an examination of the ϕ, ψ maps before and after refinement (Fig. A1) shows that a significant improvement was obtained. The low energy of the refined structures was primarily due to good interior side-chain packing, which resulted in a van der Waals' contribution to the total energy of -207 kcal. This can be clearly seen in the stereo view of the final energy-minimized structure (Fig. A2). Five new backbone hydrogen bonds

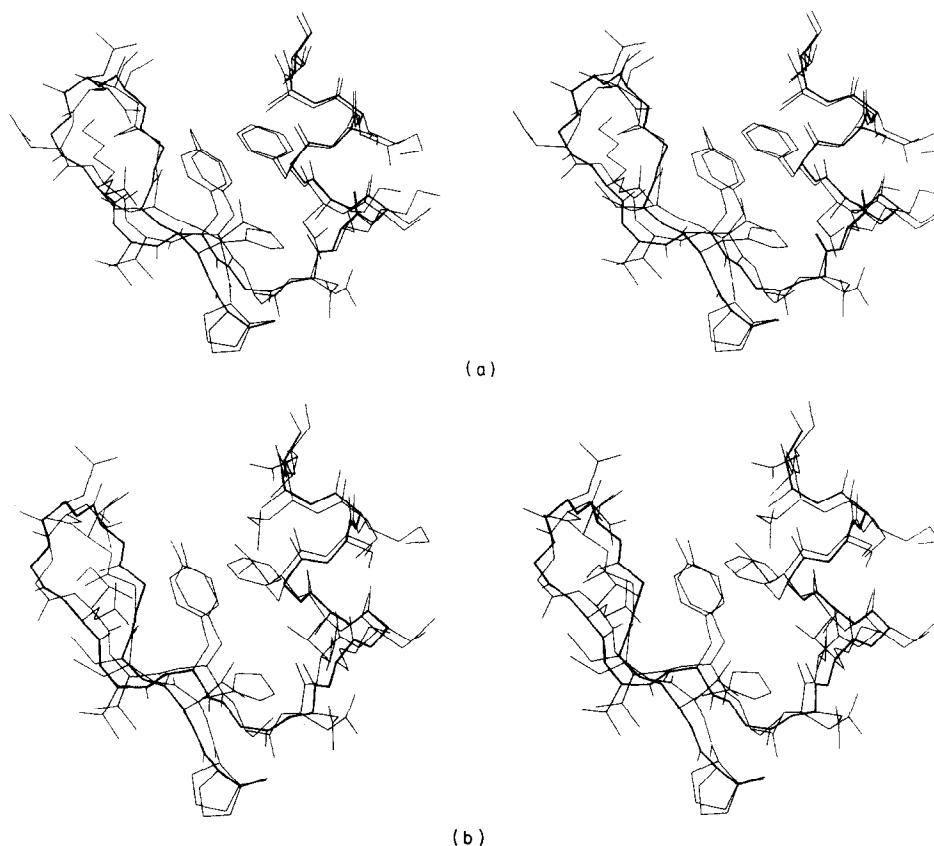


Figure A3. Stereo pairs showing the superpositions of the segment of residues from 23 to 42 in the energy-minimized structure C5 and the homologous segments of (a) the Japanese quail ovomucoid, and (b) the porcine pancreatic secretory inhibitor. The bonds connecting the backbone atoms of the BUSI IIA structure have been drawn with a heavier line. The segments have been aligned so as to minimize the RMSD in their α -carbons.

formed as a result of the energy minimization, which are given here as (carbonyl oxygen, amide proton) pairs: (Cys7, Phe10), (Ala8, Lys11), (Ala41, Ser45), (Cys25, Lys52), (Pro23, Gly55). Although the hydroxyl of Tyr32 does not titrate to a pH of at least 12 (Štrop & Wüthrich, 1983), it did not form a hydrogen bond on energy minimization. However, the side-chain oxygens of Asn28, Glu30 and Ser45 were all less than 6.0 Å away from the Tyr32 hydroxyl oxygen atom.

The RMSD between corresponding α -carbons in the BUSI structure and the homologues, PSTI and OMJPQ3, decreased during the refinement of the C5 structure (Table A1). A remarkable degree of similarity was found in the spatial locations of corresponding side-chains in the fragment of the energy-minimized BUSI structure from residues 23 to 42, and residues 22 to 41 of the homologous proteins OMJPQ3 and PSTI (Fig. A3). The RMSD between all heavy atoms in residues 23 to 42 before and after the refinement was only 0.98 Å. This shows that in the interior of the C5 structure computed by the DISGEO program directly from the n.m.r. data, the side-chains were already in approximately the correct spatial positions†, even though there were substantial uncertainties in the χ dihedral angles.

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References

- Anil Kumar, Ernst, R. R. & Wüthrich, K. (1980). *Biochem. Biophys. Res. Commun.* **95**, 1–6.
- Anil Kumar, Wagner, G., Ernst, R. R. & Wüthrich, K. (1981). *J. Amer. Chem. Soc.* **103**, 3654–3658.
- Arseniev, A. S., Kondakov, V. I., Maiorov, V. N., Volkova, T. M., Grishin, E. V., Bystrov, V. F. & Ovchinnikov, Yu. A. (1983). *Bioorgan. Khim.* **9**, 768–793.
- Billeter, M., Braun, W. & Wüthrich, K. (1982). *J. Mol. Biol.* **155**, 321–346.
- Bodenhausen, G., Vold, R. L. & Vold, R. R. (1980). *J. Magn. Reson.* **37**, 93–106.
- Bolognesi, M., Gatti, G., Menegatti, E., Guarneri, M., Marquart, M., Papamokos, E. & Huber, R. (1982). *J. Mol. Biol.* **162**, 839–868.
- Braun, W., Bösch, C., Brown, L. R., Gö, N. & Wüthrich, K. (1981). *Biochim. Biophys. Acta*, **667**, 377–396.
- Braun, W., Wider, G., Lee, K. H. & Wüthrich, K. (1983). *J. Mol. Biol.* **169**, 921–948.
- Bundi, A. & Wüthrich, K. (1979). *Biopolymers*, **18**, 285–298.
- Čechová, D., Jonáková, V., Sedláková, E. & Mach, O. (1979). *Hoppe-Seyler's Z. Physiol. Chem.* **360**, 1753–1758.
- Ebina, S. & Wüthrich, K. (1984). *J. Mol. Biol.* **179**, 283–288.
- Ferrige, A. G. & Lindon, J. C. (1978). *J. Magn. Reson.* **31**, 337–340.
- Fitzwater, S. & Scheraga, H. A. (1982). *Proc. Nat. Acad. Sci., U.S.A.* **79**, 2133–2137.
- Fritz, H., Schiessler, H., Geiger, R., Ohlsson, K. & Hochstrasser, K. (1978). *Agents Act.* **8**, 57–64.
- Fujinaga, M., Read, R. J., Sielecki, A., Ardelt, W., Laskowski, M. Jr & James, M. N. G. (1982). *Proc. Nat. Acad. Sci., U.S.A.* **79**, 4868–4872.
- Gordon, S. L. & Wüthrich, K. (1978). *J. Amer. Chem. Soc.* **100**, 7094–7096.
- Havel, T. F. & Wüthrich, K. (1984). *Bull. Math. Biol.* **46**, 673–698.
- Havel, T. F. & Wüthrich, K. (1985). *J. Mol. Biol.* **182**, 281–294.
- Havel, T. F., Crippen, G. M. & Kuntz, I. D. (1979). *Biopolymers*, **18**, 73–81.
- Havel, T. F., Kuntz, I. D. & Crippen, G. M. (1983). *Bull. Math. Biol.* **45**, 665–720.
- Kalk, A. & Berendsen, H. J. C. (1976). *J. Magn. Reson.* **24**, 343–366.
- Kato, I., Schrode, J., Wilson, K. A. & Laskowski, M. Jr (1976). *Prot. Biol. Fluids*, **23**, 235–243.
- Langridge, R., Ferrin, T., Kuntz, I. & Connolly, M. (1982). *Science*, **211**, 661–666.
- Levitt, M. (1983). *J. Mol. Biol.* **170**, 723–764.
- Luenberger, D. G. (1973). *Introduction to Linear and Nonlinear Programming*, Addison-Wesley Publishing Co., Reading, Mass.
- Macura, S., Huang, Y., Suter, D. & Ernst, R. R. (1981). *J. Magn. Reson.* **43**, 259–281.
- Marion, D. & Wüthrich, K. (1983). *Biochem. Biophys. Res. Commun.* **113**, 967–974.
- Nagayama, K. & Wüthrich, K. (1981). *Eur. J. Biochem.* **115**, 653–657.
- Némethy, G., Pottle, M. & Scheraga, H. (1983). *J. Phys. Chem.* **87**, 1883–1887.
- Nyburg, S. C. (1974). *Acta Crystallogr. sect. B*, **30**, 251–253.
- Papamokos, E., Weber, E., Bode, W., Huber, R., Empie, M. W., Kato, I. & Laskowski, M. Jr (1982). *J. Mol. Biol.* **158**, 515–537.
- Pardi, A., Billeter, M. & Wüthrich, K. (1985). *J. Mol. Biol.* In the press.
- Redfield, A. G. & Kunz, S. D. (1975). *J. Magn. Reson.* **19**, 250–254.
- Schiessler, H., Arnold, M., Ohlsson, K. & Fritz, H. (1976). *Hoppe-Seyler's Z. Physiol. Chem.* **357**, 1251–1260.
- Štrop, P. & Wüthrich, K. (1983). *J. Mol. Biol.* **166**, 631–640.
- Štrop, P., Wider, G. & Wüthrich, K. (1983). *J. Mol. Biol.* **166**, 641–667.
- Wagner, G., Anil Kumar & Wüthrich, K. (1981). *Eur. J. Biochem.* **114**, 375–384.
- Wider, G., Hosur, R. V. & Wüthrich, K. (1983). *J. Magn. Reson.* **52**, 130–135.
- Williamson, M. P., Marion, D. & Wüthrich, K. (1984). *J. Mol. Biol.* **173**, 341–360.
- Wüthrich, K. (1976). *NMR in Biological Research: Peptides and Proteins*, North-Holland Publishing Company, Amsterdam.

† During the interactive computer graphics manipulations of step 2 above, only the following dihedral angles in the segment from residues 23 to 42 were changed: χ^2 of His24, 55° to 35°; χ^1 and χ^2 of Asn34, –49° to 0° and 126° to 70°; χ^4 of Lys35, 14° to 180°. The dihedral angles changed during the step 5 were: χ^1 of Thr31, –61° to –75°; χ^1 of Cys39, –163° to –85°.

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- Wüthrich, K., Billeter, M. & Braun, W. (1983). *J. Mol. Biol.* **169**, 949–961.
- Wüthrich, K., Štrop, P., Ebina, S. & Williamson, M. P. (1984). *Biochem. Biophys. Res. Comm.* **122**, 1174–1178.
- Wüthrich, K., Billeter, M. & Braun, W. (1985). *J. Mol. Biol.* In the press.

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